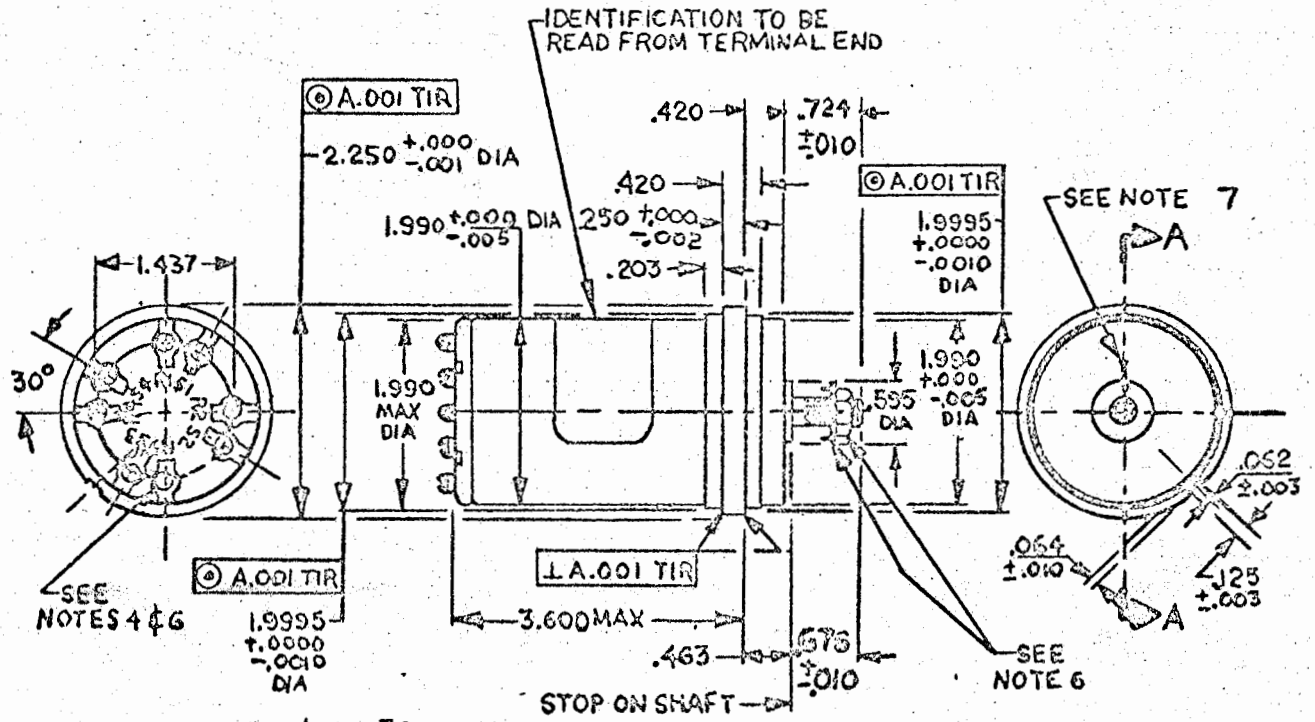
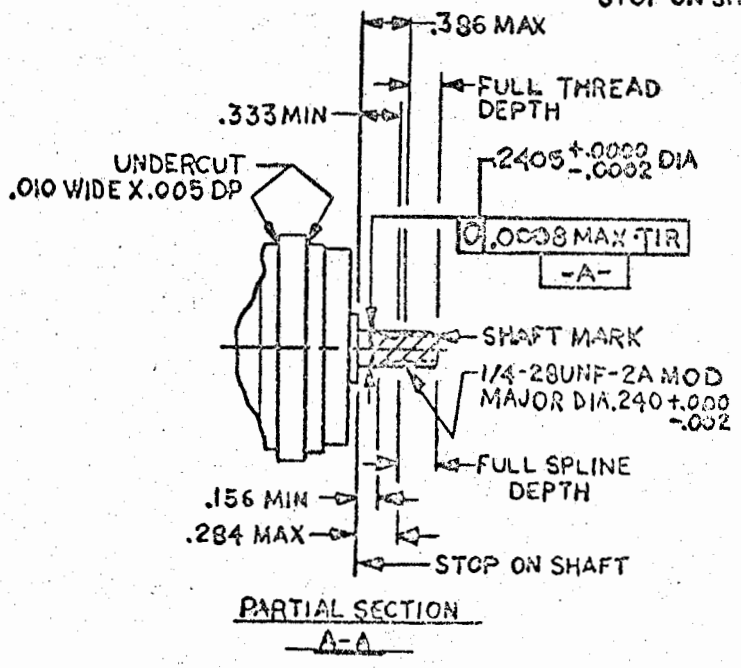
MILITARY SPECIFICATION SHEET

RESOLVER, ELECTRICAL

TYPE 23R58N4a

The complete requirements for procuring the resolver described herein shall consist of this document and the issue in effect of MIL-R-23417.




SPLINE DATA

| | |
|---|---|
| TEETH | 22 |
| PITCH | 96 |
| PITCH DIAMETER | .2291+.000-.002 |
| OUTSIDE DIAMETER | .2405+.0000-.0005 |
| PRESSURE ANGLE | 20° |
| ROOT DIA. MAX | .205 |
| TOOTH FORM | FULL DEPTH INVOLUTE |

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE BUORD929471.



FSC 5990

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NOTES:

1. Dimensions are in inches.
2. Unless otherwise specified, tolerances on decimals are .XXX ±.005 and angles ±2°.
3. Perpendicularity and concentricity shall be measured with the unit in a vertical position supported by the shaft, then the housing rotated.
4. Terminal sequence shall be as shown and identified by molded lettering on the end cap or permanently affixed thereto. The end cap and terminal block shall be secured by screws other than the terminal screws. Mounting screw positions optional. Terminal block designed for MS17182 terminal lug. (An approved 12-terminal block may be used provided all unused terminal positions are closed.)
5. Minor variations of unit configuration are permitted for undimensioned design detail.
6. The following list of hardware shall be packaged in a ventilated envelope made of material having a PH value of 6-8 and compatable with the weight of the contents. The packaged hardware shall be placed in the unit resolver package as follows:

| | | |
|---|---|---|
| Machine screw | (MS35275-213) | 8 required (*) (**) |
| Lockwasher | (MS35338-135) | 8 required (*) (**) |
| Drive washer | (MS17186-8) | 1 required (*) |
| Nut | (MS17187-3) | 1 required (*) |

(*) For units with spline shaft and terminal board termination.
(**) For units with plain smooth shaft and terminal board termination.
For information only: Use mounting clamp MS17183 or equal.

7. Mark on housing is to match shaft mark within 10° when rotor is set at approx. resolver zero, and will bear no relation to the center lines shown on the drawing.
8. A voltage value of 1 to 100 V shall be entered in the 'Volts' designation space on the identification plate.
9. With the resolver energized under standard test conditions the resonant frequency will be recorded during qualification as that point where the output voltage has a maximum value and phase shift -90°. Low corner frequency (referenced as design information) is the point where output voltage is down 3db and phase shift 45°.
10. Pending issuance of a qualified products list (QPL) a sample of the resolvers to be furnished under this specification shall be subjected to the preproduction tests as specified in MIL-R-23417.



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TABLE I

RESOLVER, ELECTRICAL: TYPE 23R58N4a

PERFORMANCE REQUIREMENTS AND CHARACTERISTICS

| | Requirements | Value | Unit | Tolerance |
|---|---|---|---|---|
| 1. | Excitation winding | Stator | | |
| 2. | Rated frequency | 400 | Hz | ±1% |
| 3. | Input voltage | | volts (rms) | ±1% |
| | Maximum | 100 | | |
| | Test | 50 | | |
| | Minimum | 1.0 | | |
| 4. | Primary current | 8.5 | milliamps | nominal |
| 5. | Primary power | 0.063 | watts | nominal |
| 6. | D. C. resistance | | ohms | ±15% |
| | Rotor | 230 | | |
| | Stator | 130 | | |
| 7. | Impedance | | ohms | R±15%+jx±10% |
| | $Z_{RO}$ | 950+j5900 | | |
| | $Z_{SO}$ | 870+j5790 | | |
| | $Z_{RS}$ | 360+j528 | | nominal |
| 8. | Transformation ratio | | | |
| | Output/input | 0.975 | | ±1% |
| 9. | T. R. variation | | % | |
| | Due to voltage | 0.05 | | maximum |
| 10. | Equality of transformation ratios | | % spread | |
| | Output/input | 0.08 | | |
| 11. | Phase shift | | degrees | |
| | Output/input | 1.2 | | ±0.5 |
| 12. | Phase shift variation | | degrees | |
| | Due to voltage | ±0.5 | | |
| 13. | Interaxis error | ± 3 | minutes | |
| 14. | Functional error | ±0.05 | % | |
| 15. | Null voltage | | mV(rms) output | maximum |
| | Fundamental | 20 | | |
| | Total | 30 | | |
| 16. | Shift of resolver zero | | minutes | |
| | With voltage | ±1.0 | | |
| | With frequency | ±0.5 | | |
| 17. | Harmonic distortion | 0.1 | % | maximum |
| 18. | Temperature rise | 20 | °C | maximum |
| 19. | Friction torque | 0.2 | oz. - in. | maximum |
| | @ -55°C | 0.4 | | maximum |



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TABLE I (Continued)

| Requirements | Value | Unit | Tolerance |
|---|---|---|---|
| 20. Radial play | 0.0010 (for reversal of 1/2 lb load) | inches | maximum |
| 21. End play | 0.0030 (for reversal of 1 lb load) | inches | ±0.002 |
| 22. Resonant frequency (see note 9) | 30 | kHz | minimum |
| 23. Low corner frequency (see note 9) | 8.3 | Hz | maximum |

Part number: The part number consists of the basic number of this specification sheet and a dash number taken from Table II.

Example: M23417/19-01

Basic document number

Dash number taken from Table II

TABLE II

| Dash No. | Shaft diameter | Shaft style | Termination | Shaft length | |
|---|---|---|---|---|---|
| | | | | Free length - L (±.010) | Dim to stop on shaft - L' (±.010) |
| -01 | .2405 | Spline | Terminal block | .724 | .676 |



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Custodians:
Army - MU
Navy - AS
Air Force - 11

Review activities:
Army - MU, EL
Navy - SH, AS, OS
Air Force - 11, 17, 85

User activities:
Army - MI, AV, ME, AT
Navy - CG, MC
Air Force - 14, 19

Preparing activity:
Army - MU

(Project 5990-0045)



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